

Hyemin Gu

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Amherst, MA 01002, USA

RESEARCH INTEREST

Mathematically Grounded Machine Learning; Generative Modeling; Stable Learning Algorithms; Scientific Applications of Machine Learning

EMPLOYMENT

• University of Massachusetts Amherst

Postdoctoral Researcher

Oct 2025 – Present

Amherst, MA

- Advisor: Markos A. Katsoulakis
- Research supported by DARPA's *The Right Space (TRS)* Program, developing mathematically grounded, stable machine learning architectures and a digital twin for supply-chain networks.
- Collaborating with researchers at Harvard Medical School and Dana-Farber Cancer Institute on mathematically grounded generative modeling for single-cell dynamics, leading the mathematical and algorithmic development of continuous trajectory inference methods.
- Collaborating with the UMass research team on Wasserstein gradient flow-based generative modeling, contributing to algorithmic development and numerical validation across multiple research directions.

• Ewha Womans University Seoul Hospital

Statistics Specialist

Jul 2020 – Dec 2020

Seoul, South Korea

- Developed an R-based Bioconductor pipeline (e.g., limma, edgeR, TCGAabiolinks) for gene expression analysis, including survival analysis and pathway enrichment
- Authored a tutorial book on reproducible gene expression workflows for biomedical researchers [[R/Bioconductor tutorial](#)]
- Conducted statistical training sessions for clinicians, enhancing research reproducibility and data literacy

EDUCATION

• University of Massachusetts Amherst

Ph.D. in Applied Mathematics

Sep 2020 - Feb 2026

Amherst, MA

- Advisor: Markos A. Katsoulakis
- Dissertation: [Robust Generative Modeling with Wasserstein Proximal Regularization](#)

• Ewha Womans University

M.S. in Applied Mathematics

Mar 2018 - Feb 2020

Seoul, South Korea

- Thesis: Convolutional Neural Network for 2D Flow Estimation Problem

• Ewha Womans University

B.S. in Mathematics and Computational science, Minor in Statistics

Mar 2014 - Feb 2018

Seoul, South Korea

- Thesis: Low cost training of a classification Neural Network with respect to Weight Selection

PUBLICATIONS

J=JOURNAL, P=PREPRINT

- [P.4] T Gamage, H Gu, Z Zhang, Z Chen, MA. Katsoulakis, L Rey-Bellet. (2026). [Fine-Tuning Generative Models for Extreme Events via CVaR-Penalized Wasserstein Gradient Flows](#).
- [P.3] H Gu, M Tyrrell, T Sahai, MA. Katsoulakis. (2026). [Certifying Residual Architectures from Their Primitives: A Sharp Stability Threshold](#).
- [J.3] YC Cheng, H Gu, et al. (2026). [PROFET Predicts Continuous Gene Expression Dynamics from scRNA-seq Data to Elucidate Heterogeneity of Cancer Treatment Responses](#). *Cell Systems*. DOI: <https://doi.org/10.1016/j.cels.2026.101710>
- [P.2] Z Zhang, H Gu, MA. Katsoulakis, T Sahai, et al. (2026). [ISOMORPH: A Supply Chain Digital Twin for Simulation, Dataset Generation, and Forecasting Benchmarks](#).
- [J.2] Z Chen, H Gu, MA Katsoulakis, L Rey-Bellet, W Zhu. (2025). [Robust Generative Learning with Lipschitz-Regularized \$\alpha\$ -Divergences Allows Minimal Assumptions on Target Distributions](#). *Information and Inference: A Journal of the IMA*, Volume 14, Issue 4, December 2025, iaaf028. DOI: <https://doi.org/10.1093/imaiai/iaaf028>
- [P.1] H Gu, MA Katsoulakis, L Rey-Bellet, BJ Zhang. (2024). [Combining Wasserstein-1 and Wasserstein-2 proximals: robust manifold learning via well-posed generative flows](#).
- [J.1] H Gu, P Birmpa, Y Pantazis, L Rey-Bellet, MA Katsoulakis. (2024). [Lipschitz-Regularized Gradient Flows and Generative Particle Algorithms for High-Dimensional Scarce Data](#). *SIAM Journal of Mathematics of Data Science*, 6 (4), 1205-1235. DOI: <https://doi.org/10.1137/23M1587841>

RESEARCH EXPERIENCE

• University of Massachusetts Amherst

Postdoctoral Research

Oct 2025 – Present

Amherst, MA

◦ Primary Research: **Stable Neural Architectures and Supply-Chain Digital Twins**

- * Developed a principled methodology for a priori assessing and certifying training stability in residual neural architectures from architectural primitives, using forward state-growth and backward sensitivity-growth bounds.
- * Developed the mathematical exposition of ISOMORPH, a Markov-chain-based digital twin of a multi-echelon logistics network, designed forecasting benchmarks, and built an [interactive demonstration platform](#) for stress testing the supply-chain network under different scenarios.

◦ Major Collaboration: **Gradient Flow-Based Generative Modeling for Single-Cell Dynamics**

- * Led the mathematical and algorithmic development of PROFET, a gradient flow-based generative model that transports probability measures to reconstruct continuous single-cell dynamics from sparsely sampled time-series scRNA-seq data, achieving 2.6–12.5× lower inference error at held-out intermediate time points than 10 state-of-the-art methods.

◦ Research Collaboration: **Fine-Tuning Generative Models for Extreme Events**

- * Contributed to the development of a CVaR-penalized Wasserstein gradient flow framework for fine-tuning generative models toward extreme-event regimes; developed the initial algorithmic implementation and refined the method for more efficient optimization of one-dimensional objectives.

• University of Massachusetts Amherst

Doctoral Research

Feb 2022 – Sep 2025

Amherst, MA

- Primary Research: **Robust Generative Modeling with Wasserstein Proximal Regularization**
 - * Developed gradient flow-based generative algorithms based on sequential particle transport using Wasserstein-1-regularized f -divergences, and extended the framework to latent-space dynamics for scalable high-dimensional learning.
 - * Developed well-posed flow-based generative models using Wasserstein-1 and Wasserstein-2 proximal-regularized f -divergences for robust learning of distributions supported on low-dimensional manifolds.
- Research Validation: **Cross-Architecture Benchmarking of Wasserstein-Proximal Regularization**
 - * Implemented and benchmarked Wasserstein-1 and Wasserstein-2 proximal regularization across diffusion models, normalizing flows, and GANs, demonstrating the data-agnostic robustness of Wasserstein-1 proximal regularization.
- Research Collaboration: **Gradient Flow-Based Generative Modeling for Single-Cell Dynamics**
 - * Developed the mathematical formulation and core algorithmic framework of PROFET, a gradient flow-based generative model for continuous trajectory inference from sparse scRNA-seq data.

INVITED TALKS

- **Robust Mamba-Based TSF via Decomposed Lipschitz Regularization** at *SIAM Conference on Uncertainty Quantification (UQ)* in Minneapolis, MN, Mar 2026
- **Generative Particle Flows with Force-Matching for Reconstructing Nonlinear Cellular Dynamics** at *SIAM NNP Sectional Annual Meeting* in University Park, PA, Nov 2025
- **Combining Wasserstein-1 and Wasserstein-2 proximals: robust manifold learning via well-posed generative flows** at *Joint Mathematics Meetings – MAA* in Seattle, WA, Jan 2025
- **Wasserstein Proximals Stabilize Training of Generative Models and Learn Manifolds** at *Mathematics of Data Science – SIAM* in Atlanta, GA, Oct 2024

POSTER PRESENTATIONS

- **Data-Driven Force-Matching and Generative Particle Flows for Reconstructing Nonlinear Dynamics from Sparse scRNA-Seq Time Series** at *Sampling, Inference, and Data-Driven Physical Modeling in Scientific Machine Learning – IPAM* in Los Angeles, CA, Jul 2025
- **Lipschitz Regularized Gradient Flows and Latent Generative Particles** at *Mathematics of Data Science – SIAM* in Atlanta, GA, Oct 2024
- **Combining Wasserstein-1 and Wasserstein-2 proximals: robust manifold learning via well-posed generative flows** at *Robust Optimization and Simulation of Complex Stochastic Systems – ICERM* in Providence, RI, Sep 2024
- **Lipschitz Regularized Gradient Flows and Latent Generative Particles** at *Optimal Transport in Data Science – ICERM* in Providence, RI, May 2023
- **Training a 2 layer Neural Network using SVD-generated weights** at *Joint Mathematics Meetings – MAA* in San Diego, CA, Jan 2018
- **Necessary and sufficient conditions for shortest vectors in lattices of dimension 2 and 3** at *Joint Mathematics Meetings – MAA* in Atlanta, GA, Jan 2017

HONORS AND AWARDS

- **Distinguished Thesis Award** Apr 2026
Department of Mathematics and Statistics, University of Massachusetts
- **Graduate Student Travel Grants** Jan 2025
American Mathematical Society
- **Anne and Peter Costa Graduate Prize in Applied Mathematics** Apr 2024
Department of Mathematics and Statistics, University of Massachusetts

PROFESSIONAL SERVICE

- **Journal Reviewer**. *SIAM Journal on Scientific Computing (SISC)*, 2026

TEACHING EXPERIENCE

- **University of Massachusetts Amherst** *Feb 2021 – Dec 2024*
Amherst, MA
Graduate Teaching Assistant
 - (MATH456) **Mathematical Modeling – Mathematics of Generative AIs**: Graded assignments for an advanced modeling course focusing on generative AI methods. (Fall 2024)
 - (MATH532H) **Nonlinear Dynamics and Chaos**: Graded assignments and led [Python tutorials for ODE solving](#). (Fall 2021)
 - (MATH545) **Linear Algebra for Applied Mathematics**: Graded assignments, held office hours, and conducted discussion sessions. (Summer 2021)
 - (MATH235) **Linear Algebra**: Graded assignments. (Spring 2021)
- **Ewha Womans University** *Mar 2018 – Dec 2019*
Seoul, South Korea
Graduate Teaching Assistant
 - **Numerical Differential Equations**: Covered ODE/PDE numerics, Monte Carlo methods, and optimization. (Fall 2019)
 - **Finite Mathematics and Programming**: MATLAB programming, logic, and combinatorics. (Spring 2019)
 - **Calculus 2**: Multivariate calculus. (Fall 2018)
 - **Numerical Analysis**: Solving linear systems, power method, numerical integration/differentiation. (Spring 2018)
 - Graded assignments and held office hours across all courses.

LEADERSHIP EXPERIENCE

- **Department seminar organizer as a part of Graduate Student Advisory Committee** *Sep 2022 - May 2025*
Department of Mathematics and Statistics, University of Massachusetts - Amherst
 - **Invited speakers** among faculties in the department for introducing their research interests to early-career graduate students and **hosted talks**.
 - **Participated in Graduate Student Advisory Committees meetings**, reported the progress, and discussed future directions.
- **Graduate Student Panelist in Open House for Accepted Students** *April 2025*
Department of Mathematics and Statistics, University of Massachusetts - Amherst
 - **Represented the graduate student body** by sharing perspectives on academic life, research, and department culture with newly admitted PhD and MS students.
- **Panelist in SIAM Graduate School Panel** *Oct 2024*
Department of Mathematics and Statistics, University of Massachusetts - Amherst
 - **Led a Q&A panel session** to introduce graduate studies in Mathematics to undergraduate mathematics majors.
- **Graduate student representative in Math Majors Fair** *April 2024*
Department of Mathematics and Statistics, University of Massachusetts - Amherst
 - **Advised undergraduate mathematics majors** on pursuing graduate studies by sharing insights into graduate life and personal research experiences.

WORKSHOPS AND TRAINING PROGRAMS

- **Industrial Mathematics Academy**

Jun 2018

National Institute for Mathematical Sciences, South Korea

- Presented final results of a group project addressing an industrial problem.
- Proposed a **Convolutional Neural Network** to classify infected individuals from medical images.
- Coordinated team efforts and participated in tutorials on **Python data analysis** and **Keras**, and lectures on **matrix-based data analysis** and **linear programming**.

- **Industrial Mathematics Academy**

Dec 2017

National Institute for Mathematical Sciences, South Korea

- Proposed a **Poisson process-based model** for assessing driving safety scores from On-board Diagnostic data.
- Attended tutorials on the **fundamentals of neural networks**.